COMP30850 Exam - Spring 2025/26

Guidelines

- Complete all three tasks. All tasks carry equal marks.
- All tasks should be completed in a single Jupyter Notebook. A template Jupyter notebook file has been provided on Brightspace.
- Use Markdown cells to briefly explain your work.
- A time limit of **2 hours** applies to this exam. You must submit your Jupyter notebook (IPYNB file) within this time.
- No collaboration is allowed. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- This is an open book exam (i.e., you can use module lecture/lab material and internet resources).
- Development environments (such as VSCode or Colab) **must not** be used for the exam. Use web-based Jupyter notebooks to complete the exam.

Use of Generative AI:

- Generative AI tools, code assistants, and large language models (such as ChatGPT, Copilot or Gemini) **must not** be used for any part of this exam.

Overview

This exam involves the analysis of a movie co-starring network derived from movies released between 2010 and 2025. The dataset captures which actors appeared together in movies, allowing exploration of collaboration patterns in the movie industry. Two CSV files are provided on Brightspace:

1. *actors.csv:* Actor names and gender, with one row per actor.
2. *movies.csv:* Movie titles and cast members, with one row per movie-actor combination.

Task 1: Network Creation

a) Load the two CSV files. From these files, create a co-starring network where actors are connected if they appeared in the same movie. Include edge weights representing the number of movies in which both actors appeared. Report the number of nodes and edges in the network.

b) Add a *gender* attribute to nodes in the network, based on the input data. Report the gender distribution of the nodes.

c) Identify the top 10 pairs of actors by edge weight.

Task 2: Community Finding

a) Apply a community finding algorithm to the co-starring network. Report the number of communities detected.

b) Calculate the proportion of nodes assigned to each community. How evenly is the network split across communities?

c) Create a subgraph corresponding to the largest community. Calculate the density of this subgraph and compare it to the density of the full network. What does this suggest about collaboration patterns within the community?

Task 3: Centrality and Ego Network Analysis

a) Identify the top 10 actors in the network by weighted degree centrality.

b) Identify the female actor with the highest weighted degree centrality. Extract the ego network for this actor and report how many nodes are in the ego network and what percentage of the full network's nodes this represents.

c) Calculate the proportion of female and male actors in the ego network. Compare this to the gender distribution of the full network.